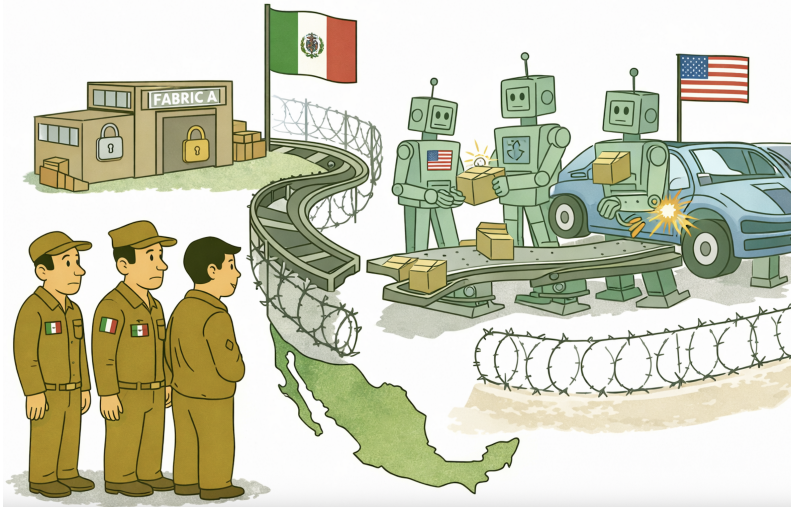
The Political Economy of Automation and Fragmented Production: Evidence from Mexico



Carles Boix, Valentina González-Rostani & Erica Owen | October 2026 | WashU

Motivation

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Reshoring

Trump Leans on Carrier to Keep 2,000 U.S. Jobs From Moving to Mexico



Reverse decision to offshore

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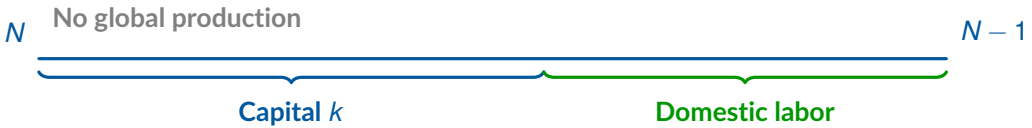
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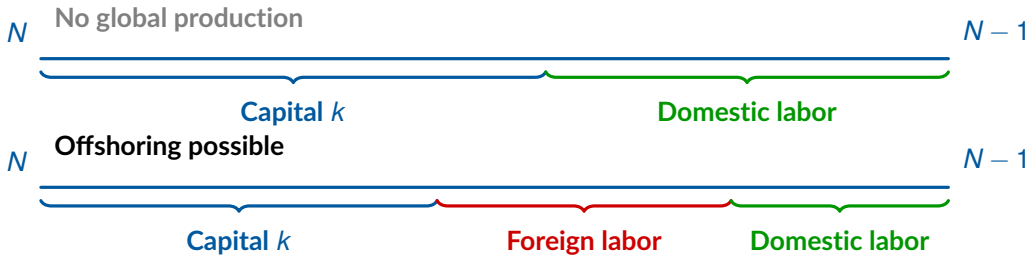
More exposed communities show:

- ↓ Wages
- ↑ Violent organized crime
- ↑ Support for left-populist presidential candidates

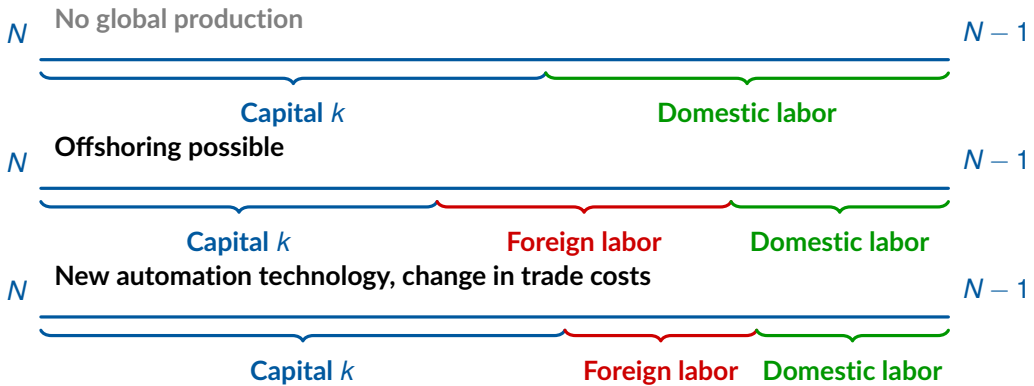
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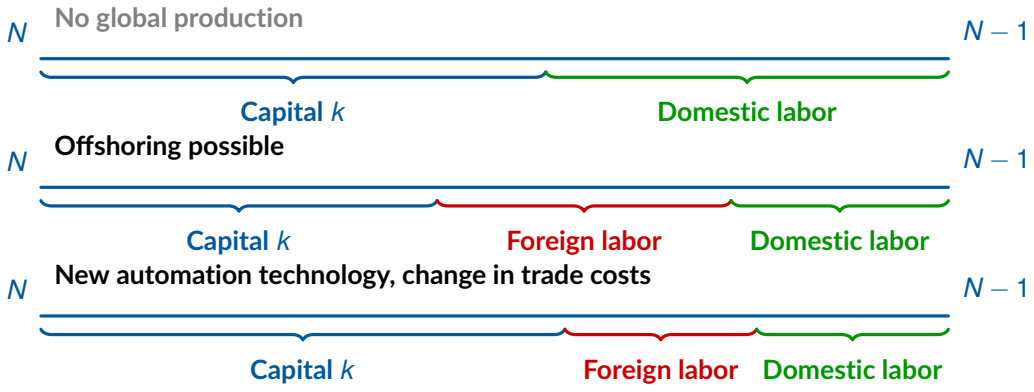
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The intuition of the tasks framework



Foreign robot exposure as negative labor market shock

Foreign robots & organized crime

- **Negative labor-market shocks** ↓ legal employment & wages (Cavazos Hernandez & Sivakumar, 2022; Dix-Carneiro et al., 2018; Dube et al., 2016)

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H1: Exposure to foreign robots increases violent organized crime

Foreign robots & populism

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Foreign robots & populism

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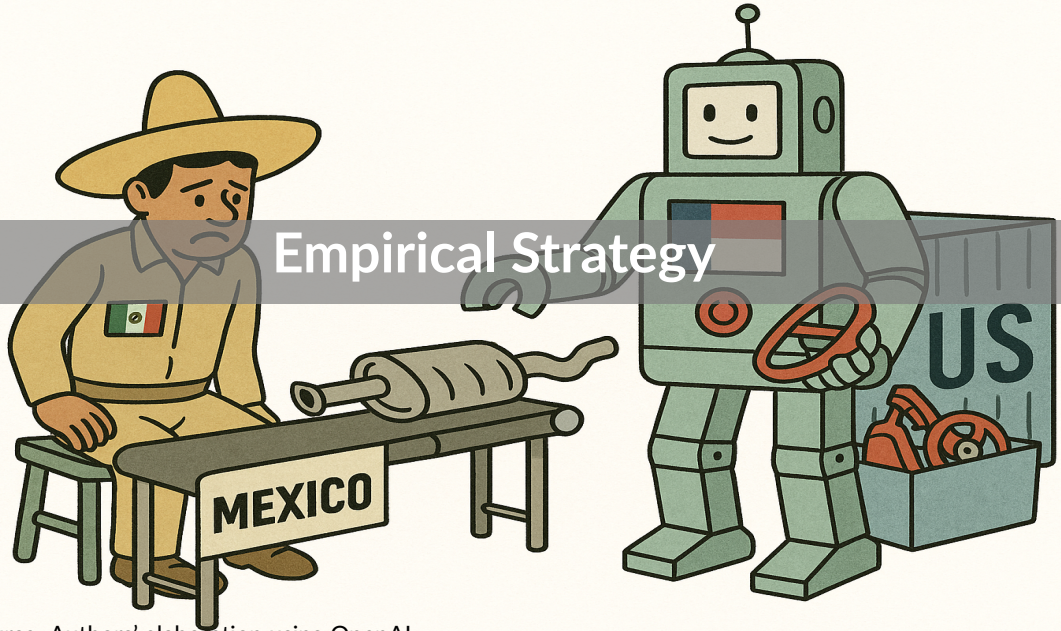
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The electoral response depends on the parties and programs available.

H2: Foreign robot exposure increases support for left-populist candidates in Mexico

Empirical Strategy



Source: Authors' elaboration using OpenAI

Sample and dependent variables

Unit of analysis: About 1,800 Mexican 🇲🇽 CZs

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Violence and organized crime

- Homicide-rate change, 1999–2018 (per 10,000; INEGI)
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Politics (2006–2024)

- Change in left vote share: AMLO (2006) to Sheinbaum (2024)
- Other major parties (PRI, PAN)

Term	President	Party
1994–2000	Zedillo	PRI
2000–2006	Fox	PAN
2006–2012	Calderón	PAN
2012–2018	Peña Nieto	PRI
2018–2024	López Obrador	Morena
2024–	Sheinbaum	Morena

Domestic robot exposure 🇲🇽 🤖

$$\text{Domestic robot exposure}_c = \sum_i \ell_{ci,1990} \frac{\Delta R_i^{MX}}{L_{i,1990}}$$

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- 1990 CZ industry employment shares ($\ell_{ci,1990}$) $\times$ growth in **Mexican** robots

$$\Delta R_i^{MX} = R_{i,2015}^{MX} - R_{i,2000}^{MX}$$

- *Source:* Robots (IFR); employment shares from Mexican Census

Foreign robot exposure 🇺🇸 🤖

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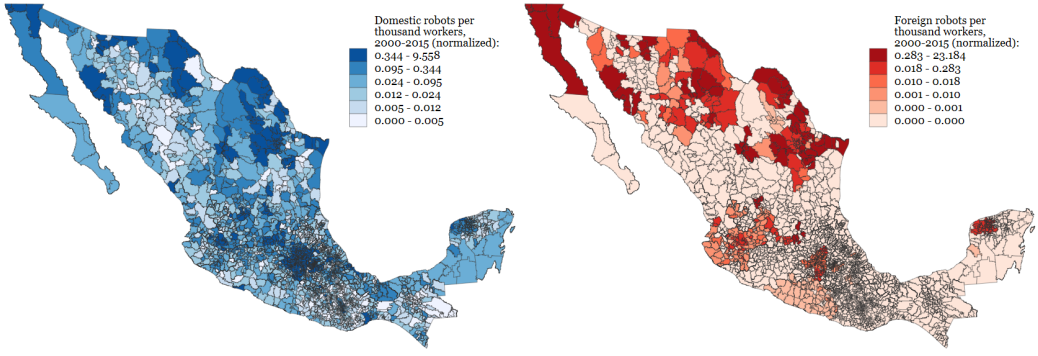
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- Normalize by employment in maquiladoras in industry i: $L^f_{i,1990}$.
- Weight by CZ's initial reliance on maquiladora employment: $\ell^f_{ci,1990}$
- *Source:* Data on maquiladora employment (CEPAL); Offshorability (UN Comtrade/BLS). (Faber, 2020)

Exposure to domestic and foreign robots, 2000–2015



Domestic Robots & Foreign Robots

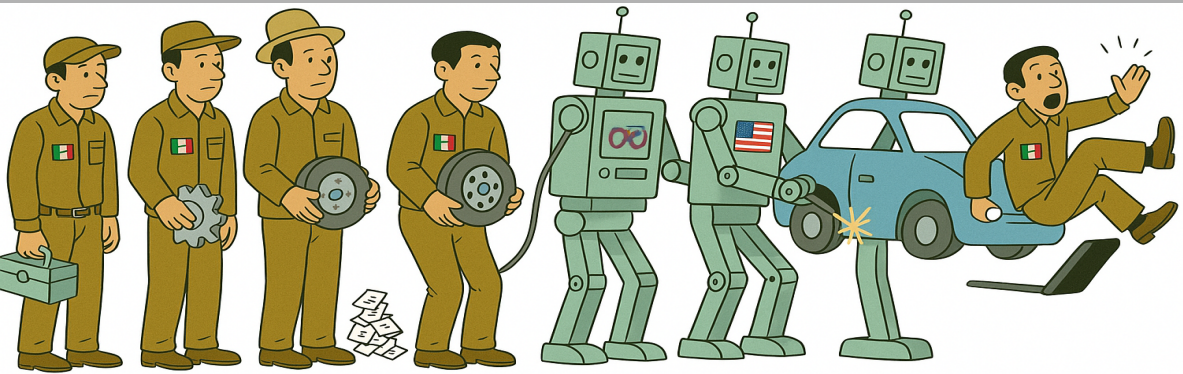
Empirical strategy

- **Shift-share IV:** Replace U.S./Mexican robot growth with European industry adoption 🌍. Use an all-origin outsourcing weight for foreign exposure.

$$\Delta Y_c = \alpha + \beta\,\text{Foreign Exposure}_c + \gamma\,\text{Domestic Exposure}_c + \mathbf{X}_c'\theta + \varepsilon_c$$

- **Identifying assumption:** European robot growth is unrelated to other shocks to Mexican CZ outcomes, conditional on controls.
- **Controls:** baseline demographics and industry composition, routine work, NAFTA exposure, and Chinese import competition.

Results



Source: Authors' elaboration using OpenAI

Mechanisms: Firms' stated reasons for reshoring and wages

Topics in reshoring news (LDA)

Label	Top 10 Representative Words	Share
Proximity and Incentives	specified, proximity, incentives, customers, market, government, labor, concessions, tax, infrastructure	0.3517
Technology and Automation	workforce, eco, synergies, availability, training, skilled, technology, automation, innovation, manufacturing	0.2461
Supply Chain and Quality	time, cost, lead, market, quality, rework, warranty, total, supply, chain	0.2027
Brand Image and Wages	image, brand, customer, wages, rising, capacity, utilized, risk, responsiveness, improvement	0.1995

Note: Data obtained from the Reshore Now library. Analysis of 184 projects from 2006-2024.

Technology and automation account for 24.61% of topic share

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Economic consequences: Foreign exposure is associated with lower wages (census data).

Foreign robot exposure and rising violence

IV	(1) Δ Homicides	(2) Δ Org. Crime
Exposure to foreign robots	20.06*	2.700***
	(11.26)	(0.946)
Exposure to domestic robots	-16.00	-3.441**
	(11.98)	(1.748)
Controls	✓	✓
Contemporary	✓	✓
Demographics	✓	✓
Industry	✓	✓
Observations	1801	1805
R^2	0.405	0.352
F	0.791	7.688
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Foreign robot exposure and greater left support

IV	(1) Δ Left (2024)	(2) Δ Right (2024)
Exposure to foreign robots	1.120***	-0.739**
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Exposure to domestic robots	-2.231	3.524
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Conclusion

- **Automation crosses borders.** 🇺🇸 🤖 can displace 🇲🇽 workers while the productive investment occurs abroad.
- **In Mexico:** Foreign exposure is associated with lower manufacturing wages, more organized crime, and greater left-populist support.
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- **Export-led development remains vulnerable.** Upgrading and diversification matter when foreign technology erodes low labor-cost advantages. 🌍🤖

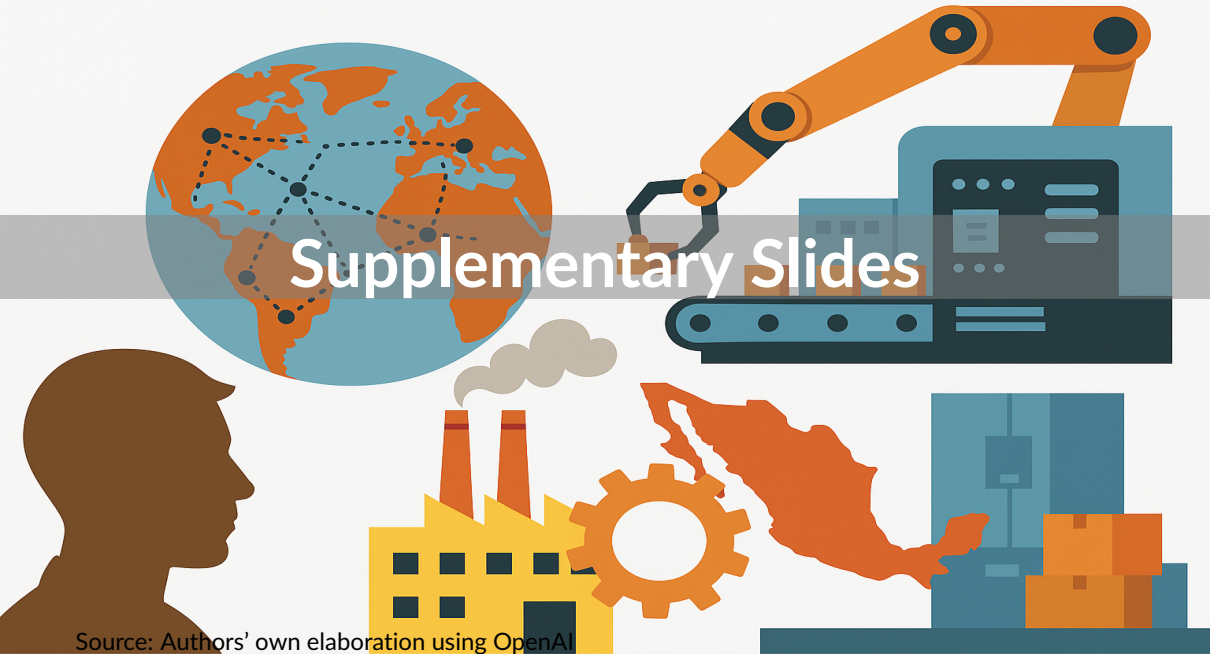
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Thank you!

Valentina González-Rostani
gonzalez-rostani.com | gonzalez.rostani@usc.edu

Supplementary Slides



Source: Authors' own elaboration using OpenAI

Appendix - Index

Moved from the main talk

- Theory
  - Literature
  - Argument
  - Firm decisions
  - Labor shock
- Context
  - U.S. robots
  - Mexico
- Evidence
  - Table 1: wages
  - Additional tests

Data and identification

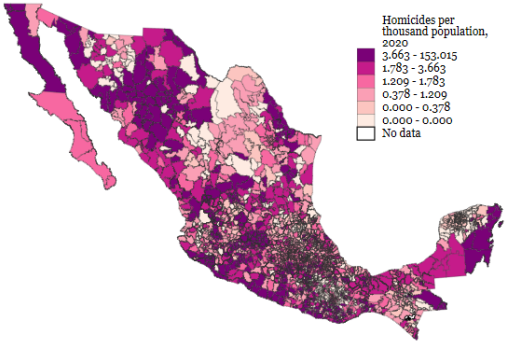
- Robot data
  - Crime data
  - Controls
- Exposure
  - IV
  - Equation
  - China

Additional analyses retained from UW

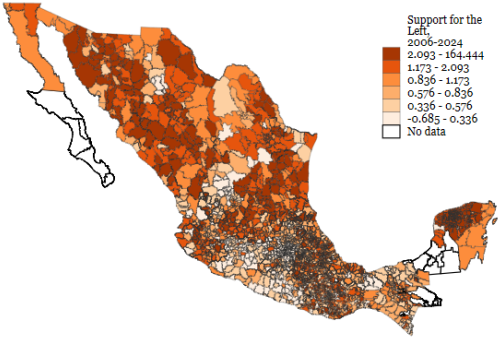
- Motivation
  - Carrier
  - Reshoring
- Results
  - Crime
  - Vote
  - Employment/exports
  - Wages
  - Industry exports
- Robustness
  - FR/DR
  - Crime 1993
  - Narcocrime
  - Vote 1993
- Candidates
  - Sheinbaum
  - Alvarez
  - Galvez Ruiz
  - Null
- Mechanisms
  - Crime
  - Despair
  - Strikes

Main talk: Overview · Tasks · Crime theory · Populism · Design · Table 3 · Table 4 · Conclusion

Commuting zone-level variation in DVs



Violent crime



Support for the Left

Literature gap & relevance

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  Boosts productivity but hurts certain workers (job loss, wage decline)
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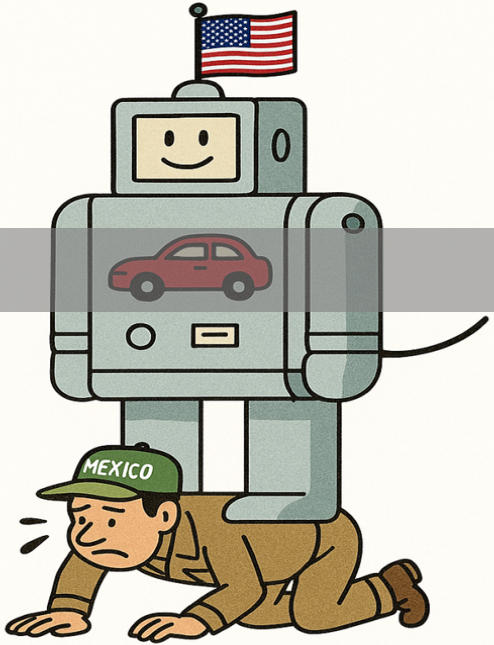
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Our focus: How Northern automation shocks transmit through GVC to reshape labor and politics in the Global South.

The Argument



Source: Authors' elaboration using OpenAI

The Economic Consequences of Automation

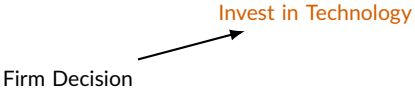
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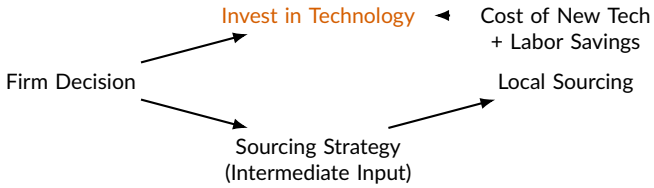
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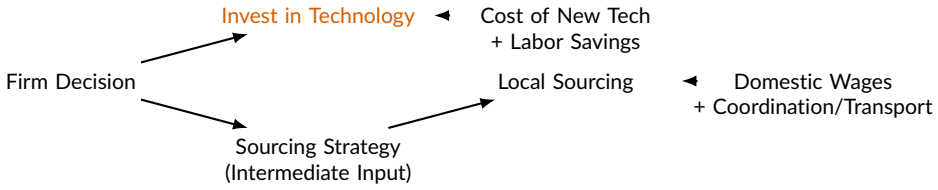
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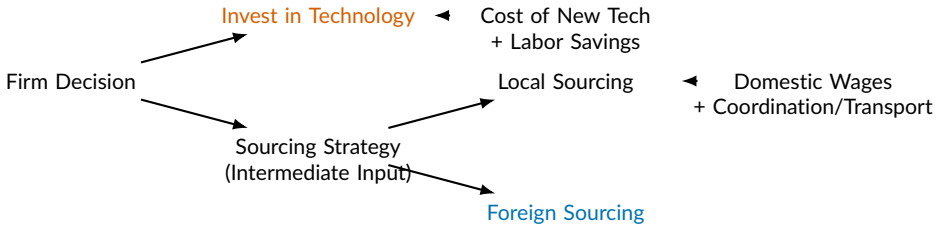
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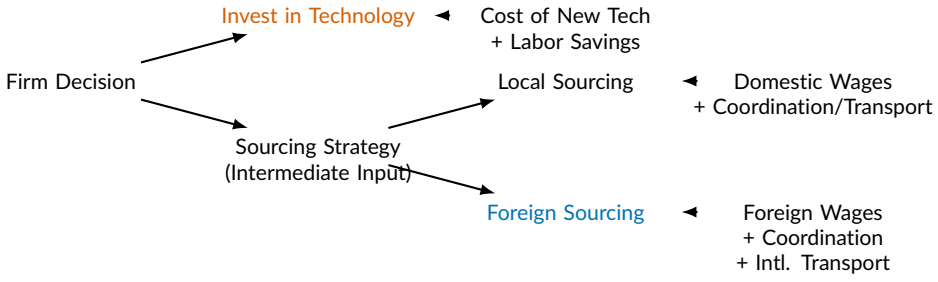
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Economic consequences of foreign automation in Global South

In Global South, automation linked to:

Economic consequences of foreign automation in Global South

In Global South, automation linked to:

- Lower employment, rising informality (Faber, 2020)
- Reshoring (Pinheiro et al., 2023)
- Mixed effect on amt. of exports, but change in composition (Artuc et al., 2022, 2019; Faber, 2020; Stemmler, 2023)

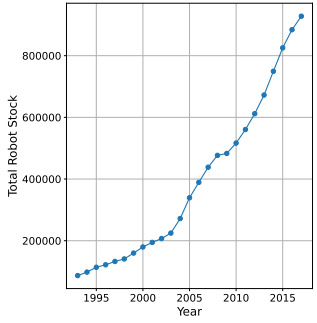
Economic consequences of foreign automation in Global South

In Global South, automation linked to:

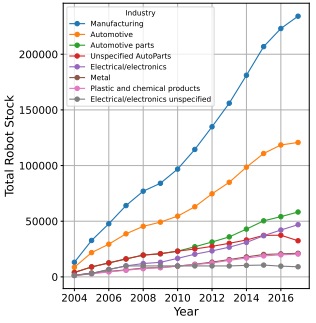
- Lower employment, rising informality (Faber, 2020)
- Reshoring (Pinheiro et al., 2023)
- Mixed effect on amt. of exports, but change in composition (Artuc et al., 2022, 2019; Faber, 2020; Stemmler, 2023)

Foreign robot exposure as negative labor market shock

Empirical context: U.S. robot adoption



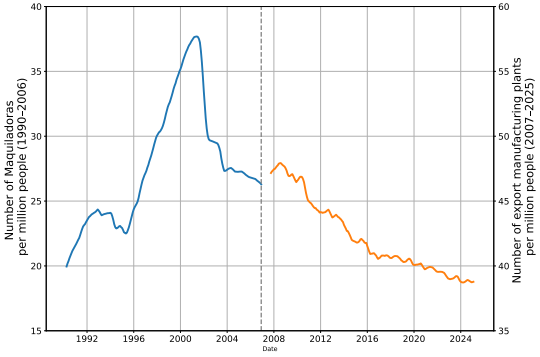
(a) Industrial robots, 1993–2017.



(b) Robots by industry, 2004–2017.

Notes: Data from IFR. Authors' calculations.

Empirical context: Mexico's manufacturing shift



Note: This figure plots the 4-month moving averages of maquiladoras and export manufacturing plants per million inhabitants in Mexico. Authors' own elaboration based on data from INEGI.

Foreign robot exposure lowers manufacturing wages

Table 1. Manufacturing wages, 2000–2015

Wages	OLS (1) Δ Wages	IV (2) Δ Wages
Exposure to foreign robots	-0.0270*	-0.0289*
	(0.0152)	(0.0152)
Exposure to domestic robots	-0.116*	-0.112*
	(0.0603)	(0.0582)
Controls	✓	✓
Demographics	✓	✓
Industry	✓	✓
Observations	1610	1610
R^2	0.014	0.014
F	8.602	8.198
Kleibergen-Paap Wald F-stat		100.2

Note: Residualized manufacturing wage changes. OLS uses external exposures; IV instruments both robot exposures. Both columns unweighted. State-clustered SEs. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

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Foreign and domestic robot exposure are associated with lower wages

Additional tests

Robustness

- Domestic exposure has different security and electoral estimates
- Controls: distance to the U.S., state FE, drug trafficking routes
- Include domestic and foreign exposure measures separately, partial instrumentation foreign exposure.

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Mechanism (Suggestive)

- Lower manufacturing wages; complementary evidence on employment and exports
- Economic channel, not increase in other crimes (e.g., no rise in property)
- Earlier supplementary analysis: exposed regions report fewer **strikes** (outside the current paper)

Reshoring reasons (Word Cloud)



Note: Analysis of 184 reshoring projects, 2006-2024

Example

Trump Leans on Carrier to Keep 2,000 U.S. Jobs From Moving to Mexico



Results- Employment & Exports

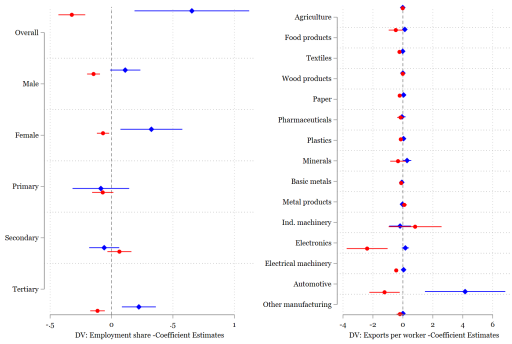


Figure: Changes in employment (left panel) and exports (right panel) in relation to exposure to domestic and foreign robots, 2000–2015. Blue coefficients indicate exposure to domestic robots; red coefficients indicate exposure to foreign robots.

Results - Wages

OLS	(1) Δ Wages	(2) Δ Employment	(3) Δ Working Age Pop.
External exposure to foreign robots	-0.0339***	-1.220***	-0.841***
	(0.0108)	(0.338)	(0.255)
External exposure to domestic robots	-0.0864*	3.068	2.532
	(0.0484)	(2.084)	(1.560)
Demographics	✓	✓	✓
Industry	✓	✓	✓
Region	✓	✓	✓
Observations	1610	1800	1800
R^2	0.030	0.245	0.388
IV	(1) Δ Wages	(2) Δ Employment	(3) Δ Working Age Pop.
Exposure to foreign robots	-0.0361***	-1.354***	-0.934***
	(0.0113)	(0.400)	(0.307)
Exposure to domestic robots	-0.0832*	2.876	2.379*
	(0.0468)	(1.938)	(1.445)
Demographics	✓	✓	✓
Industry	✓	✓	✓
Region	✓	✓	✓
Observations	1610	1800	1800
R^2	0.029	0.243	0.386
F	28.62	12.51	19.06
Kleibergen-Paap Wald F-stat	107.2	159.6	159.6

Results - Organized Crime

OLS	(1) Homicides	(2) Kidnapping	(3) Narco	(4) Human Traffic	(5) Δ Org. Crime	(6) Crimes
External exposure to foreign robots	0.211**	0.0100**	0.592**	0.00336**	2.640***	0.747
	(0.100)	(0.00388)	(0.285)	(0.00155)	(0.681)	(0.718)
External exposure to domestic robots	-0.781**	-0.0107	-0.792	0.00185	-2.662**	-7.017**
	(0.322)	(0.0189)	(0.559)	(0.00616)	(1.251)	(3.353)
Demographics	✓	✓	✓	✓	✓	✓
Industry	✓	✓	✓	✓	✓	✓
Region	✓	✓	✓	✓	✓	✓
Observations	1802	1802	1802	1802	1033	1802
R^2	0.197	0.130	0.455	0.127	0.460	0.166
IV	(1) Homicides	(2) Kidnapping	(3) Narco	(4) Human Traffic	(5) Δ Org. Crime	(6) Crimes
Exposure to foreign robots	0.234**	0.0111***	0.654*	0.00369**	2.924***	0.833
	(0.114)	(0.00397)	(0.336)	(0.00184)	(0.851)	(0.769)
Exposure to domestic robots	-0.734**	-0.00981	-0.733	0.00187	-2.455*	-6.638**
	(0.297)	(0.0175)	(0.543)	(0.00580)	(1.283)	(3.103)
Demographics	✓	✓	✓	✓	✓	✓
Industry	✓	✓	✓	✓	✓	✓
Region	✓	✓	✓	✓	✓	✓
Observations	1802	1802	1802	1802	1033	1802
R^2	0.203	0.132	0.409	0.122	0.418	0.170
F	9.267	21.68	6.305	15.00	8.272	12.09
Kleibergen-Paap Wald F-stat	172.7	172.7	172.7	172.7	169.2	172.7

▸ Notes

Organized Crime - Notes

The dependent variable in column 1 refers to the homicide rate, while column 2 refers to the number of homicides per 10,000 population, both sourced from CONAPO. Column 3 refers to the number of narcocrimes per 10,000 population, sourced from CISEN. All specifications include the following control variables: 1) Region: fixed effects for eight broad regions in Mexico; 2) Demographics: 1990 CZ demographics, including the share of men and the share of people with primary education as their highest level; 3) Industry: shares of employment in manufacturing in 1990 and the 1990 level of employment-to-population ratio. All regressions are weighted by a CZ's share of the national working-age population in 1990. Standard errors are robust to heteroskedasticity and clustered by state. The coefficients marked with $***$, $**$, and $*$ are significant at the 1%, 5%, and 10% confidence levels, respectively. Full specifications for these additional analyses are retained from the UW deck.

Table

Results - Populism

OLS	(1) Sheinbaum (Left)	(2) Galvez (Right)	(3) Δ Left	(4) Δ Right
External exposure to foreign robots	0.00628*	-0.00741*	0.0614**	-0.0119**
	(0.00340)	(0.00386)	(0.0224)	(0.00487)
External exposure to domestic robots	0.000131	-0.00393	0.0404	-0.00299
	(0.00919)	(0.00968)	(0.0839)	(0.0146)
Demographics	✓	✓	✓	✓
Industry	✓	✓	✓	✓
Region	✓	✓	✓	✓
Observations	1800	1800	1792	1792
R^2	0.677	0.581	0.335	0.579
IV	(1) Sheinbaum (Left)	(2) Galvez (Right)	(3) Δ Left	(4) Δ Right
Exposure to foreign robots	0.00685*	-0.00809*	0.0671***	-0.0130**
	(0.00370)	(0.00417)	(0.0245)	(0.00528)
Exposure to domestic robots	0.000627	-0.00434	0.0434	-0.00380
	(0.00836)	(0.00884)	(0.0790)	(0.0134)
Demographics	✓	✓	✓	✓
Industry	✓	✓	✓	✓
Region	✓	✓	✓	✓
Observations	1800	1800	1792	1792
R^2	0.679	0.585	0.335	0.582
F	19.39	46.33	41.76	30.78
Kleibergen-Paap Wald F-stat	384.1	384.1	384.0	384.0

▸ Notes

Populism - Notes

The dependent variables in columns 1–2 represent each candidate's share of valid votes for the 2024 elections. Columns 3–4 capture the change in support between 2006 and 2024. For the left-bloc comparison, we contrast support for AMLO in the 2006 presidential election—when he ran under the left coalition *Coalición por el Bien de Todos* (PRD–PT–Convergencia)—with support for Sheinbaum in 2024, who ran under *MORENA* in alliance with *Sigamos Haciendo Historia*. For the right-bloc comparison, we contrast support for Gálvez Ruiz in 2024 with the main right-of-center candidates in 2006: Felipe Calderón, who ran under the *PAN*, and Roberto Madrazo, who ran under the *Alianza por México* (PRI–PVEM). We also include Roberto Campa of *Nueva Alianza*, whose platform in 2006 leaned toward the center-right. All specifications include the following control variables: 1) Region: fixed effects for eight broad regions in Mexico; 2) Demographics: 1990 CZ demographics, including the share of men and the share of people with primary education as their highest level; 3) Industry: shares of employment in manufacturing in 1990 and the 1990 level of employment-to-population ratio. All regressions are weighted by a CZ's share of the national working-age population in 1990. Standard errors are robust to heteroskedasticity and clustered by state. The coefficients marked with $***$, $**$, and $*$ are significant at the 1%, 5%, and 10% confidence levels, respectively. Full specifications for these additional analyses are retained from the UW deck.

◂ Table

Correlation b/t domestic and foreign

Table: Correlation Matrix: Exposure to Domestic and Foreign Robots (2000–2015)

	Domestic Robots (DR)	Foreign Robots (FR)
Domestic Robots (DR)	1.0000	
Foreign Robots (FR)	0.4189	1.0000

Robustness- Narcocrime

IV	(1) Homicides	(2) Kidnapping	(3) Narco	(4) Human Traffic	(5) Δ Org. Crime
Exposure to foreign robots	0.234**	0.0111***	0.654*	0.00369**	2.924***
	(0.114)	(0.00397)	(0.336)	(0.00184)	(0.851)
Exposure to domestic robots	-0.734**	-0.00981	-0.733	0.00187	-2.455*
	(0.297)	(0.0175)	(0.543)	(0.00580)	(1.283)
Demographics	✓	✓	✓	✓	✓
Industry	✓	✓	✓	✓	✓
Region	✓	✓	✓	✓	✓
Observations	1802	1802	1802	1802	1033
R^2	0.203	0.132	0.409	0.122	0.418
F	9.267	21.68	6.305	15.00	8.272
Kleibergen-Paap Wald F-stat	172.7	172.7	172.7	172.7	169.2

Note: Columns 1-5 are estimated in levels, with DV measured post robot shock. Column 6 represents the change in organized crime between 2000 and 2018.

↑ Foreign robot exposure → ↑ Violent organized crime

Robustness: Property Crimes

Table: Effect of Robot Exposure on tasa_2018 (2000–2015)

	(1) DR only	(2) FR only	(3) DR + FR (IV)
Exposure to domestic robots	−6.310**		−6.702**
	(2.990)		(3.053)
Exposure to foreign robots		0.619	0.835
		(0.836)	(0.769)

Robust SEs clustered by state. Weighted by 1990 pop.

Robustness: Homicide rate

Table: Effect of Robot Exposure on hom_2018_pop (2000–2015)

	(1) DR only	(2) FR only	(3) DR + FR (IV)
Exposure to domestic robots	−0.642**		−0.754**
	(0.254)		(0.294)
Exposure to foreign robots		0.210	0.235**
		(0.128)	(0.114)

Robust SEs clustered by state. Weighted by 1990 pop.

Robustness: Kidnapping

Table: Effect of Robot Exposure on secuestro_pop (2000–2015)

	(1) DR only	(2) FR only	(3) DR + FR (IV)
Exposure to domestic robots	−0.005		−0.009
	(0.015)		(0.017)
Exposure to foreign robots		0.011***	0.011***
		(0.004)	(0.004)

Robust SEs clustered by state. Weighted by 1990 pop.

Robustness: Narcocrime

Table: Effect of Robot Exposure on narco_pop (2000–2015)

	(1) DR only	(2) FR only	(3) DR + FR (IV)
Exposure to domestic robots	−0.475		−0.670
	(0.443)		(0.519)
Exposure to foreign robots		0.630*	0.652*
		(0.345)	(0.337)

Robust SEs clustered by state. Weighted by 1990 pop.

Robustness: Human trafficking

Table: Effect of Robot Exposure on trata_personas_pop (2000–2015)

	(1) DR only	(2) FR only	(3) DR + FR (IV)
Exposure to domestic robots	0.003		0.002
	(0.005)		(0.006)
Exposure to foreign robots		0.004**	0.004**
		(0.002)	(0.002)

Robust SEs clustered by state. Weighted by 1990 pop.

Robustness: Sheinbaum

Table: Effect of Robot Exposure on sheinbaum (2000–2015)

	(1) DR only	(2) FR only	(3) DR + FR (IV)
Exposure to domestic robots	0.022*		0.020*
	(0.012)		(0.011)
Exposure to foreign robots		0.009***	0.008***
		(0.003)	(0.003)

Robust SEs clustered by state. Weighted by 1990 pop.

Robustness: Alvarez

Table: Effect of Robot Exposure on alvarez (2000–2015)

	(1) DR only	(2) FR only	(3) DR + FR (IV)
Exposure to domestic robots	−0.004		−0.004
	(0.005)		(0.004)
Exposure to foreign robots		−0.003*	−0.003*
		(0.002)	(0.002)

Robust SEs clustered by state. Weighted by 1990 pop.

Robustness: Galvez Ruiz

Table: Effect of Robot Exposure on galvezruiz (2000–2015)

	(1) DR only	(2) FR only	(3) DR + FR (IV)
Exposure to domestic robots	−0.018		−0.016
	(0.013)		(0.012)
Exposure to foreign robots		−0.005*	−0.005*
		(0.003)	(0.003)

Robust SEs clustered by state. Weighted by 1990 pop.

Robustness: Null votes

Table: Effect of Robot Exposure on nulos (2000–2015)

	(1) DR only	(2) FR only	(3) DR + FR (IV)
Exposure to domestic robots	0.001		0.001
	(0.001)		(0.001)
Exposure to foreign robots		0.000	0.000
		(0.000)	(0.000)

Robust SEs clustered by state. Weighted by 1990 pop.

Robustness: Not Death of Despair

	(1) Sexual Crime	(2) Family Violence	(3) Property Crimes
Exposure to domestic robots	-0.234	-1.529	0.357
	(0.157)	(1.017)	(1.420)
Exposure to foreign robots	0.00658	-0.0426	-0.113
	(0.0472)	(0.521)	(0.390)
Share of routine workers in 1990	0.845	-10.90	24.14
	(3.346)	(20.54)	(16.15)
Exposure to Chinese import competition	0.161**	1.000***	0.460
	(0.0594)	(0.361)	(0.723)
Exposure to tariff changes from NAFTA	12.20	144.0*	-1.928
	(15.23)	(82.60)	(67.68)
Change in employment-to-population ratio 00-15	-0.00856	-0.000150	0.145*
	(0.0157)	(0.0934)	(0.0794)
Demographics	✓	✓	✓
Industry	✓	✓	✓
Region	✓	✓	✓
Observations	1802	1802	1802
R^2	0.499	0.587	0.522

Table: Impact of exposure to robots on other crimes associated with the Death of Despair hypothesis (OLS).

▸ Notes

Other Crimes - Notes

The dependent variable in column 1 refers to the homicide rate, while column 2 refers to the number of these types of crimes per 10,000 population, both sourced from CONAPO. All specifications include the following control variables: 1) Region: fixed effects for eight broad regions in Mexico; 2) Demographics: 1990 CZ demographics, including the share of men and the share of people with primary education as their highest level; 3) Industry: shares of employment in manufacturing in 1990 and the 1990 level of employment-to-population ratio. All regressions are weighted by a CZ's share of the national working-age population in 1990. Standard errors are robust to heteroskedasticity and clustered by state. The coefficients marked with $***$, $**$, and $*$ are significant at the 1%, 5%, and 10% confidence levels, respectively.

Table

Robustness: 1993 - 2015 Organized Crime

Table: Effects of Robot Exposure (1993-2015) on Violence

OLS	(1) Crimes	(2) Homicides	(3) Kidnapping	(4) Narco	(5) Human Traffic	(6) Δ Org. Crime
External exposure to domestic robots	-11.71*	-1.199**	-0.0288*	-0.217	0.00402	-4.842**
	(5.985)	(0.564)	(0.0169)	(0.743)	(0.00979)	(1.950)
External exposure to foreign robots	0.427	0.164**	0.00600*	0.436**	0.00231**	2.010***
	(0.433)	(0.0698)	(0.00335)	(0.202)	(0.000885)	(0.450)
Demographics	✓	✓	✓	✓	✓	✓
Industry	✓	✓	✓	✓	✓	✓
Region	✓	✓	✓	✓	✓	✓
Observations	1802	1802	1802	1802	1802	1033
R^2	0.173	0.204	0.130	0.469	0.128	0.454
IV	(1) Crimes	(2) Homicides	(3) Kidnapping	(4) Narco	(5) Human Traffic	(6) Δ Org. Crime
Exposure to domestic robots	-9.763*	-0.953**	-0.0221	-0.0122	0.00432	-3.278*
	(5.139)	(0.466)	(0.0137)	(0.643)	(0.00795)	(1.728)
Exposure to foreign robots	0.621	0.188**	0.00666*	0.455**	0.00234**	2.165***
	(0.394)	(0.0731)	(0.00343)	(0.223)	(0.00103)	(0.509)
Demographics	✓	✓	✓	✓	✓	✓
Industry	✓	✓	✓	✓	✓	✓
Region	✓	✓	✓	✓	✓	✓
Observations	1802	1802	1802	1802	1802	1033
R^2	0.173	0.207	0.130	0.428	0.126	0.422
F	11.03	7.613	28.34	9.956	12.47	8.005
Kleibergen-Paap Wald F-stat	438.6	438.6	438.6	438.6	438.6	418.6

Additional analysis: Strikes

	(1) # Strikes	(2) # Strikes
Exposure to domestic robots	-0.00605	-0.00622
	(0.0206)	(0.0194)
Exposure to foreign robots	-0.0115**	-0.0129**
	(0.00493)	(0.00546)
Pre-Shock Strikes (1999)	0.105***	0.105***
	(0.00108)	(0.00109)
Share of routine workers in 1990	1.071***	1.087***
	(0.375)	(0.372)
Exposure to Chinese import	-0.00281	-0.000691
	(0.00693)	(0.00737)
Exposure(NAFTA)	3.985**	3.995**
	(1.699)	(1.685)
Change in employment-to-pop	-0.0162***	-0.0162***
	(0.00299)	(0.00297)
Demographics	✓	✓
Industry	✓	✓
Region	✓	✓
Observations	1805	1805
R^2	0.950	0.950
F	1894.5	1896.5
Kleibergen-Paap Wald F-stat		14284.4

Table: Earlier supplementary analysis of robot exposure and strikes.

▸ Notes

Strikes - Notes

The dependent variable is the number of strikes initiated (huelgas estalladas, INEGI). The variation refers to the period 1999 and 2016. Column (1) reports OLS estimates, and Column (2) reports IV estimates. All specifications include the following controls: (i) Region: fixed effects for eight broad regions in Mexico; (ii) Demographics: 1990 CZ demographics, including the share of men and the share of individuals with only primary education; (iii) Industry: 1990 employment shares in manufacturing and the 1990 employment-to-population ratio. All regressions are weighted by a CZ's share of the national working-age population in 1990. Standard errors are robust to heteroskedasticity. The coefficients marked with $***$, $**$, and $*$ are significant at the 1%, 5%, and 10% confidence levels, respectively.

◂ Table

Robustness: 1993 - 2015 Populism Support

OLS	(1) Sheinbaum (Left)	(2) Galvez (Right)	(3) Δ Left	(4) Δ Right
External exposure to domestic robots	-0.00141	-0.00124	0.0249	-0.00976
	(0.0176)	(0.0183)	(0.211)	(0.0229)
External exposure to foreign robots	0.00520*	-0.00546*	0.0238	-0.00846**
	(0.00287)	(0.00274)	(0.0181)	(0.00346)
Demographics	✓	✓	✓	✓
Industry	✓	✓	✓	✓
Region	✓	✓	✓	✓
Observations	1800	1800	1792	1792
R^2	0.677	0.581	0.331	0.581
IV	(1) Sheinbaum (Left)	(2) Galvez (Right)	(3) Δ Left	(4) Δ Right
Exposure to domestic robots	0.00180	-0.00422	0.0351	-0.0132
	(0.0141)	(0.0142)	(0.181)	(0.0174)
Exposure to foreign robots	0.00542*	-0.00566**	0.0244	-0.00866**
	(0.00296)	(0.00284)	(0.0179)	(0.00362)
Demographics	✓	✓	✓	✓
Industry	✓	✓	✓	✓
Region	✓	✓	✓	✓
Observations	1800	1800	1792	1792
R^2	0.679	0.585	0.330	0.583
F	173.1	95.30	7.836	26.70
Kleibergen-Paap Wald F-stat	331.5	331.5	331.3	331.3

Empirical Strategy

The empirical strategy follows the work of Acemoglu and Restrepo (2020) and Faber (2020). First, the exposure to domestic robots is measured as:

$$\text{Exposure to domestic robots}_{c(t_0,t_1)} = \sum_{i \in I} \ell_{ci,1990} \left(\frac{R^{MX}_{i,t_1} - R^{MX}_{i,t_0}}{L_{i,1990}} \right)$$

where R^{MX}_{i,t_1} and R^{MX}_{i,t_0} represent the number of robots in industry i at time t_1 and t_0 in Mexico, respectively, while $\ell_{ci,1990}$ is the share of employment in industry i out of total employment in the region c in 1990, and $L_{i,1990}$ is the total employment in industry i in 1990.

$$\text{Exposure to foreign robots}_{c(t_0,t_1)} = \sum_{i \in I} \ell^{f}_{ci,1990} \left(\frac{\left(R^{US}_{i,t_1} - R^{US}_{i,t_0} \right) O_{i,1992}}{L^{f}_{i,1990}} \right)$$

where R^{US}_{i,t_1} and R^{US}_{i,t_0} are the estimated number of robots in industry i at times t_1 and t_0 in the US, respectively, $\ell^{f}_{ci,1990}$ is the share of export-producing employment in industry i out of total CZ in 1990, $L^{f}_{i,1990}$ is the total foreign employment in industry i, and $O_{i,1992}$ is the initial share of inputs into industry-good i that are offshorable.

Shift-Share

We address potential endogeneity arising from the correlation between robot adoption and unobserved factors affecting local labor markets by employing an instrumental variable approach, using the increase in robots in the rest of the world as an instrument for robot adoption in Mexico.

$$\text{External exposure to domestic robots}_{c(t_0,t_1)} \equiv \sum_{i \in I} \ell_{ci,1990} \left(\frac{R^{WLD}_{i,t_1} - R^{WLD}_{i,t_0}}{L_{i,1990}} \right)$$

$$\text{External exposure to foreign robots}_{c(t_0,t_1)} \equiv \sum_{i \in I} \ell^{f}_{ci,1990} \left(\frac{\left(R^{WLD}_{i,t_1} - R^{WLD}_{i,t_0} \right) \hat{O}_{i,1990}}{L^{f}_{i,1990}} \right)$$

The superscript WLD denotes the sum over European countries that are also incorporating technology (i.e, excluding the US and Mexico) for which industry-level data are available from 1993 onward. To address potential endogeneity in our initial offshoring to Mexico proxy, we follow Feenstra and Hanson (1999) and Faber (2020) in defining it as the share of imported intermediate inputs from the same industry over total non-energy intermediates in U.S. industry *i* in 1990 (across all source countries).

Equation

The equation we will estimate is as follows:

$$\Delta \text{Y}_{c(t_1)} = \alpha + \beta^{d} \text{Exp. to domestic robots}_{c(t_0,t_1)} + \beta^{f} \text{Exp. to foreign robots}_{c(t_0,t_1)} + \mathbf{X}_{c,t_0}\gamma + \varepsilon_{c(t_0,t_1)}$$

Data: Controls – Economic and Demographic Context

CZ Employment & Occupation Characteristics (Faber 2020):

- Routine task share (based on U.S. crosswalk from Autor (2013)).
- Manufacturing share and employment-to-population ratio (1990).

NAFTA Exposure:

- Based on 1990 employment shares and industry tariff changes from NAFTA.
- Computed as $\sum_i \ell_{ci,1990} \Delta\tau_i$

Chinese Import Exposure:

- Bartik-style measure using Chinese imports to Mexico and the U.S.
- Includes indirect effects via offshoring to U.S.

Demographics:

- Share male and with only primary education (1990).
- Working-age population change in violence and voting models; state fixed effects in robustness checks.

Control Variables: China

$$\text{Exp. to Chinese import competition}_{c(t_0,t_1)} = \sum_{i \in I} \ell_{ci,t_0} \left[\frac{I_{i,t_1}^{CNMX} - I_{i,t_0}^{CNMX} + O_{i,t_0} \left(I_{i,t_1}^{CNUS} - I_{i,t_0}^{CNUS} \right)}{L_{i,t_0}} \right]$$

where I_{i,t_1}^{CNMX} and I_{i,t_0}^{CNMX} represent the value of imports from China to Mexico in industry i at times t_1 and t_0, respectively, and I_{i,t_1}^{CNUS} and I_{i,t_0}^{CNUS} represent the same for imports to the US. L_{i,t_0} is the total employment in industry i at time t_0, and O_{i,t_0} is the initial share of imported intermediate goods in US industry i.

Data: Independent Variable – Robot Exposure

Source: Faber (2020), using IFR data on industrial robots and Mexican Census data.

Domestic Robots:

- Bartik-style measure combining IFR robot counts and industry employment shares across CZs.
- Reflects automation based on 1990 industrial composition.

Foreign Robots (Offshoring):

- Measures U.S. offshoring intensity by industry.
- Uses UN Comtrade and BLS data; mapped from SIC72 to IFR industries.
- Weighted by Maquiladora employment data from CEPAL (1994).

Data: Dependent Variables – Crime

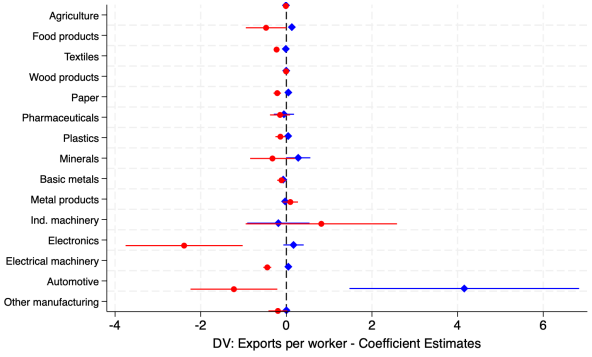
Crime Measures:

- Homicide rates per 10,000 (INEGI, 2018).
- Narcocrime incidents (CISEN data, 2015–2019).
- Crime data aggregated from municipalities to CZs.

Proxy Approach:

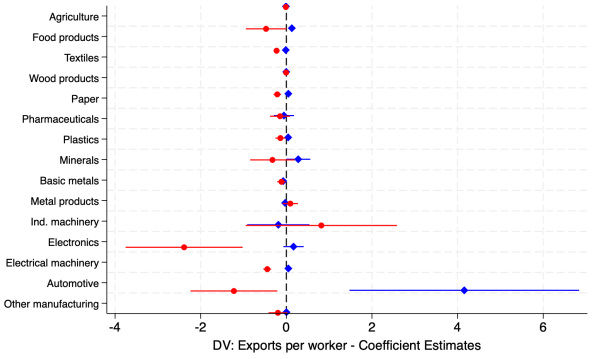
- Total homicide rate used as a proxy for organized crime, following prior literature.

Effects of robots by industry on exports per worker



Robot exposure, 2000–2015; export changes, 2004–2014
Domestic robots (blue diamonds); Foreign robots (red circles)

Effects of robots by industry on exports per worker



Robot exposure, 2000–2015; export changes, 2004–2014
Domestic robots (blue diamonds); Foreign robots (red circles)

↑ Foreign robot exposure → ↓ Exports

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